3567. Minimum Absolute Difference in Sliding Submatrix

You are given an m x n integer matrix grid and an integer k.

For every contiguous k x k **submatrix** of grid, compute the **minimum absolute** difference between any two **distinct** values within that **submatrix**.

Return a 2D array ans of size (m - k + 1) x (n - k + 1), where ans[i][j] is the minimum absolute difference in the submatrix whose top-left corner is (i, j) in grid.

Note: If all elements in the submatrix have the same value, the answer will be 0.

A submatrix (x1, y1, x2, y2) is a matrix that is formed by choosing all cells matrix[x][y] where x1 <= x <= x2 and y1 <= y <= y2.

Example 1:

Input: grid = [[1,8],[3,-2]], k = 2

Output: [[2]]

Explanation:

- There is only one possible k x k submatrix: [[1, 8], [3, -2]].
- Distinct values in the submatrix are [1, 8, 3, -2].
- The minimum absolute difference in the submatrix is |1 - 3| = 2. Thus, the answer is [[2]].

Example 2:

Input: grid = [[3,-1]], k = 1

Output: [[0,0]]

Explanation:

- Both k x k submatrix has only one distinct element.
- Thus, the answer is [[0, 0]].

Example 3:

Input: grid = [[1,-2,3],[2,3,5]], k = 2

Output: [[1,2]]

Explanation:

- There are two possible k × k submatrix:
  - Starting at (0, 0): [[1, -2], [2, 3]].
    - Distinct values in the submatrix are [1, -2, 2, 3].
    - The minimum absolute difference in the submatrix is |1 - 2| = 1.
  - Starting at (0, 1): [[-2, 3], [3, 5]].
    - Distinct values in the submatrix are [-2, 3, 5].
    - The minimum absolute difference in the submatrix is |3 - 5| = 2.
- Thus, the answer is [[1, 2]].

Constraints:

- 1 <= m == grid.length <= 30
- 1 <= n == grid[i].length <= 30
- -10^5 <= grid[i][j] <= 10^5
- 1 <= k <= min(m, n)

Solution:

```
class Solution {
public:
    vector<vector<int>> minAbsDiff(vector<vector<int>>& grid, int k) {
        int m = grid.size(), n = grid[0].size();

        vector<vector<int>> res(m - k + 1, vector<int>(n - k + 1, 0));

        for (int i = 0; i <= m - k; i++) {
            for (int j = 0; j <= n - k; j++) {
                vector<int> nums;
                for (int x = i; x < i + k; x++) {
                    for (int y = j; y < j + k; y++) {
                        nums.push_back(grid[x][y]);
                    }
                }
                sort(nums.begin(), nums.end());

                int minDiff = INT_MAX;
                for (int t = 1; t < nums.size(); t++) {
                    if (nums[t] != nums[t - 1]) {
                        minDiff = min(minDiff, nums[t] - nums[t - 1]);
                    }
                }
                if (minDiff != INT_MAX) {
                    res[i][j] = minDiff;
                }
            }
```

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        }

        return res;
    }
};
```